BSc Honours research · School of Computer Science and Applied Mathematics

A Representation-Centred Evaluation of Latent Diffusion for Afrikaans Text Generation

Hadi Butt

Supervisor: Dr Devon Jarvis

University of the Witwatersrand, Johannesburg



One passage, encoded and decoded

SOURCE · validation passage

Dié publikasie kyk na van die basiese aspekte van padveiligheid, wat spoed, dronkbestuur en **padwoede** insluit.

RECONSTRUCTION · after encoding and decoding

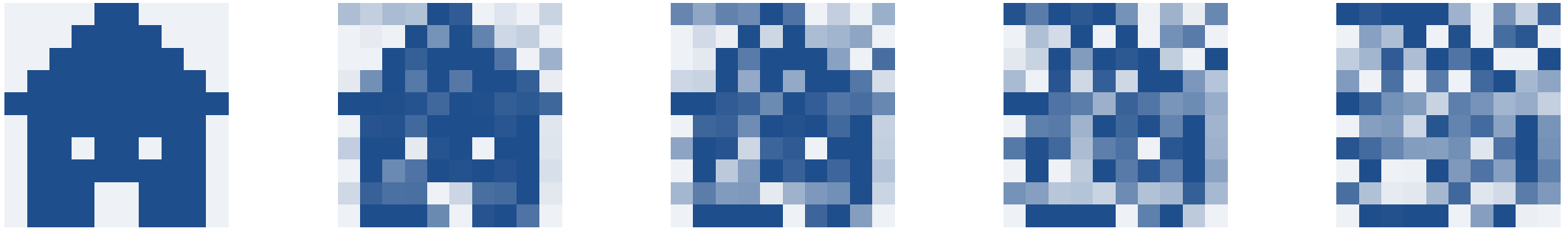
Dié publikasie kyk na van die basiese aspekte van padveiligheid, wat spoed, dronkbestuur en **padbou** insluit.

chrF 92.3 31 of 32 tokens right

padwoede → padbou road rage → road building

Diffusion learns to reverse a gradual noising process

Forward: add Gaussian noise step by step (fixed, nothing learned)



Reverse: a network learns to remove the noise, one step at a time

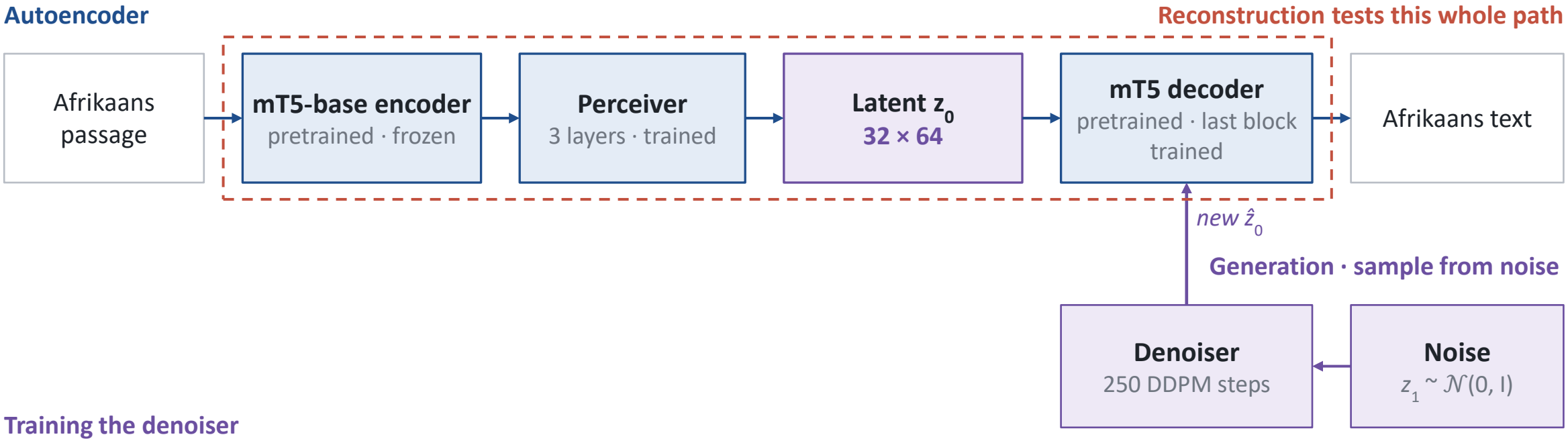
Images: continuous values

$$\text{pixel } 0.62 \;+\; \text{noise } 0.09 \;=\; 0.71$$

Text: discrete tokens

$$\text{“huis”} \;+\; \text{noise} \;=\; ?$$

Latent diffusion: generate in a continuous space, then decode



Training the denoiser

Forward noising $z_t = \sqrt{a(t)}\, z_0 + \sqrt{(1 - a(t))}\, \varepsilon, \quad a(t) = \cos^2(\pi t/2), \quad \varepsilon \sim \mathcal{N}(0, I)$

Noisy latent = mix of clean latent and Gaussian noise · clean at t = 0, pure noise at t = 1 (cosine schedule)

Training objective $\min_\theta \mathbb{E}_{z_0, t, \varepsilon} \| \hat{v}_\theta(z_t, t) - v_t \|^2, \quad v_t = \sqrt{a(t)}\, \varepsilon - \sqrt{(1 - a(t))}\, z_0$

Denoiser predicts the velocity · mean squared error of the velocity prediction

Why start with Afrikaans

DOCUMENTED CORPUS

NCHLT

Clean Afrikaans Text Corpus: government-domain text, openly licensed (CC BY 2.5 South Africa).

LIMITED LOCAL DATA

1.23 M

training words after preprocessing, from 2,005 documents.

PRETRAINED COVERAGE

mT5 · mGPT

Afrikaans is covered by both the encoder–decoder and the external scorer.

RQ1

Implementation control

What does an English control reveal before the Afrikaans results are interpreted?

RQ2

Representation fidelity

Do the Afrikaans latents keep enough information to reconstruct the text?

RQ3

Comparative generation

Does Afrikaans latent diffusion compare credibly with an autoregressive baseline?

Hypothesis: the system is credible only if it keeps the content, scores within 1.20 × an AR baseline, and doesn't degenerate.

What I trained and compared

DATA



Train · 2,005 documents

Val · Test 251 each

47,767 training passages

1,228,292 training words

≤ 50 words per passage

GENERATORS COMPARED

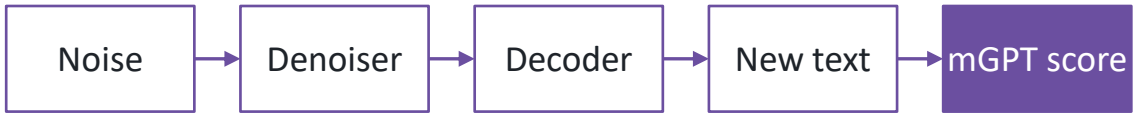
Small diffusion denoiser
8.1 M parameters · 5 seeds (42–46)

Large diffusion denoiser
188 M parameters · 5 seeds (42–46)

GPT-style AR baseline
67 M parameters · 1 seed (42)
random initialisation · no mT5 weights

Two kinds of evaluation, two different questions

Generation



How probable does another model find the generated text?

- mGPT perplexity (lower = more probable to mGPT)
- Diversity, repetition, copying of training text

Measures familiarity, not correctness.

Reconstruction



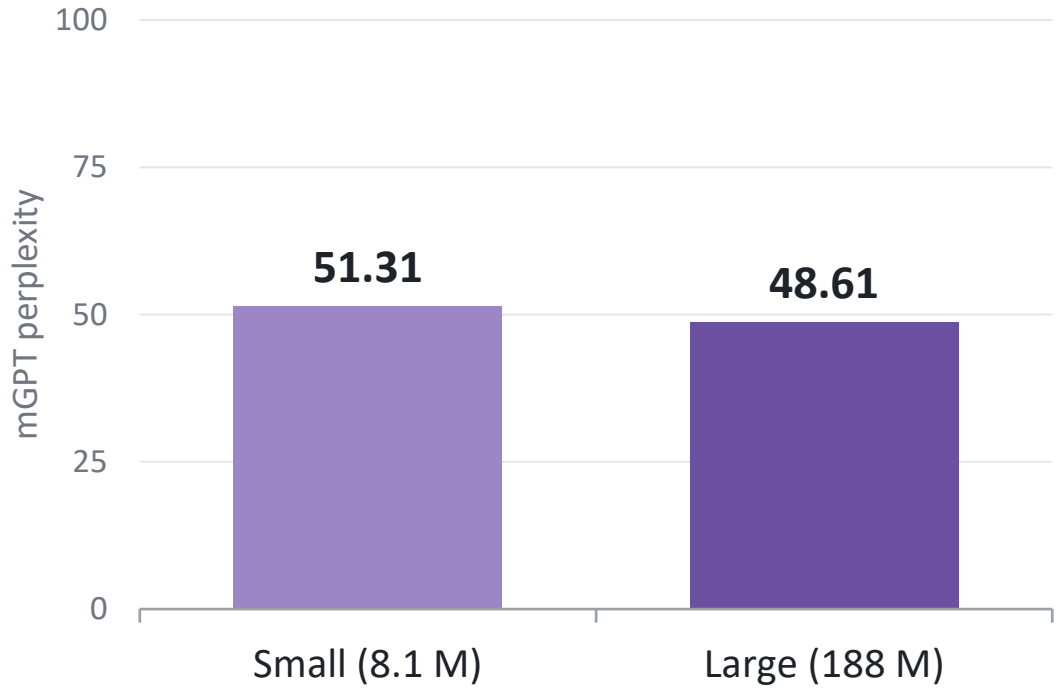
How faithfully does the original passage survive?

- chrF, exact matches, numeral recall

Compared with the source passage.

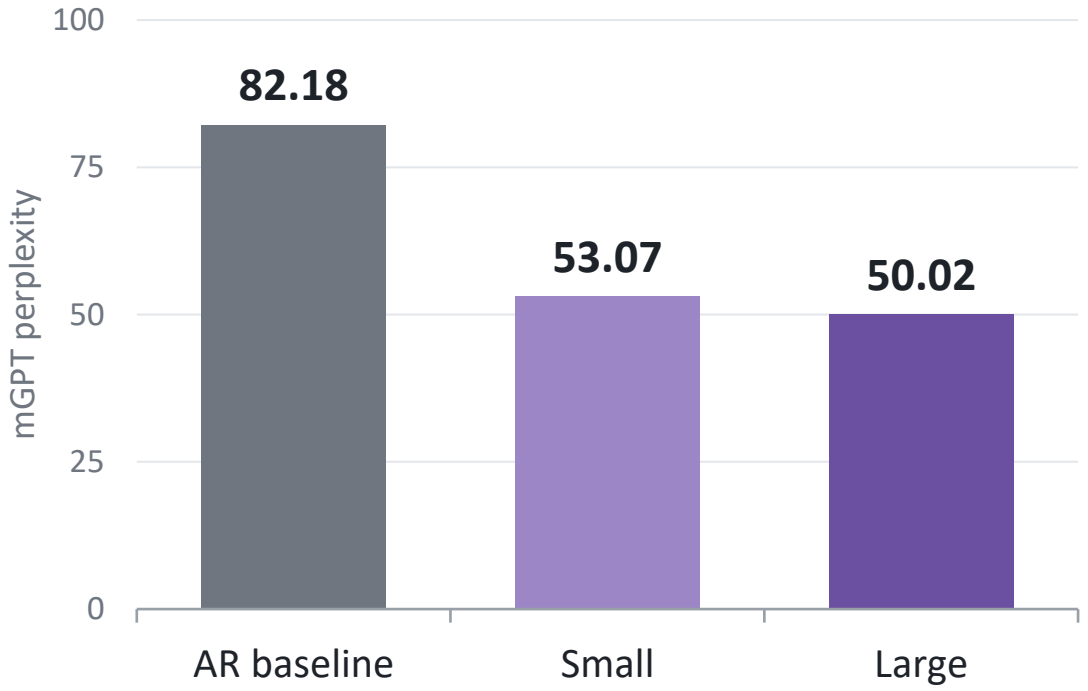
Generation scores look favourable

A · Denoiser size test split · 5 seeds × 1,000 samples



5% lower perplexity for the large denoiser

B · Against the AR baseline validation · 256 per system



Ratios to AR: 0.65 and 0.61 (limit 1.20)

Real test passages score 33.62: a reference point, not a model.

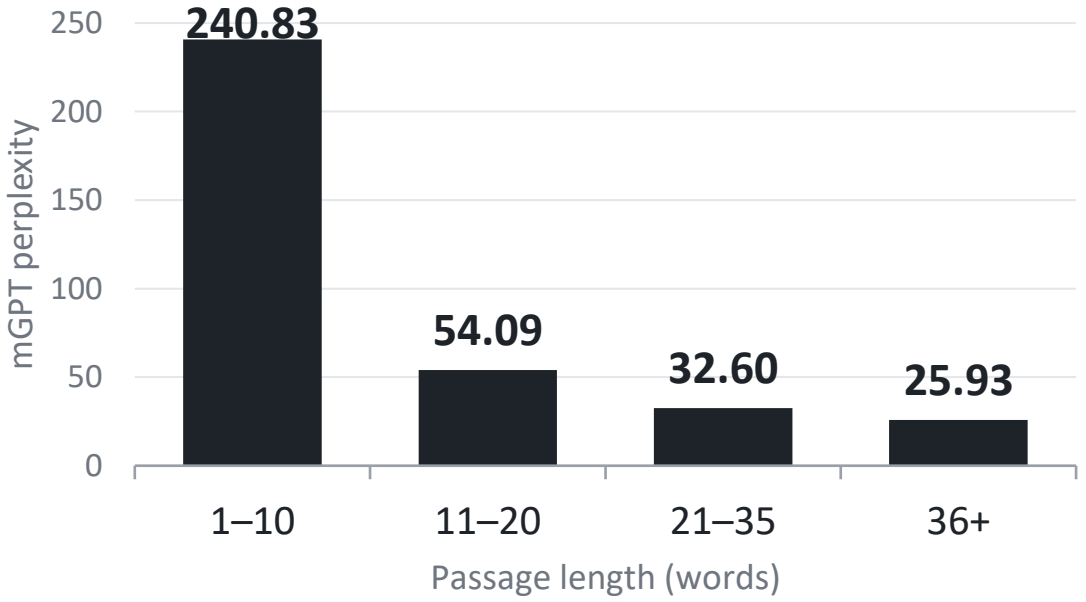
Two reasons not to read this as better Afrikaans

1 · The systems are not matched

	Diffusion	AR baseline
Pretrained weights	adapted mT5	**none**
Text decoding	beam 4, constrained	nucleus, p = 0.95
Precision	FP32	BF16
Training time	2.9–7.9 h	4.5 h

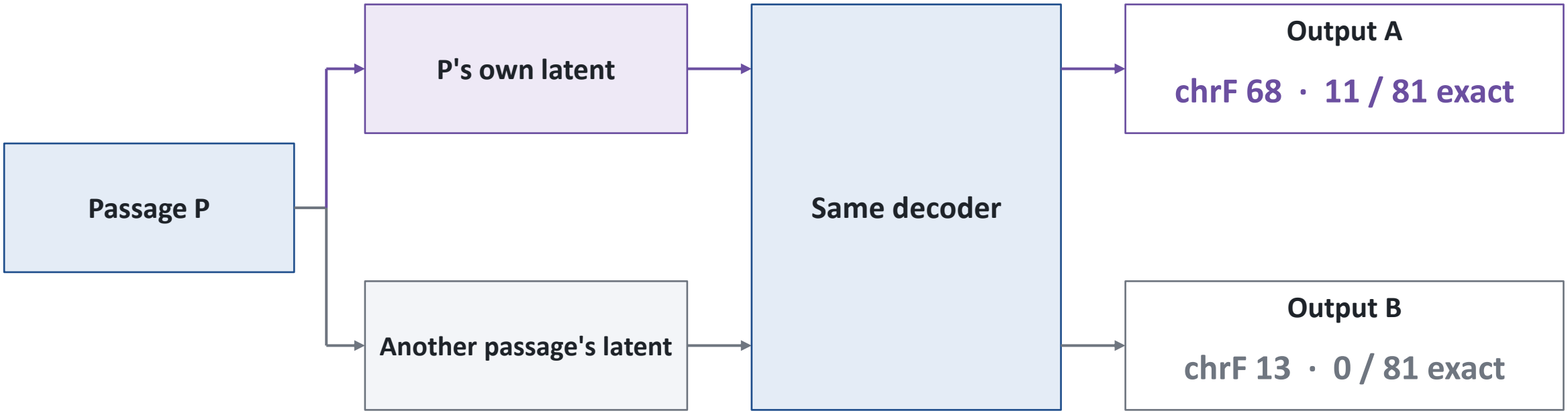
Two systems are compared, not two generation methods.

2 · The scorer is length-sensitive



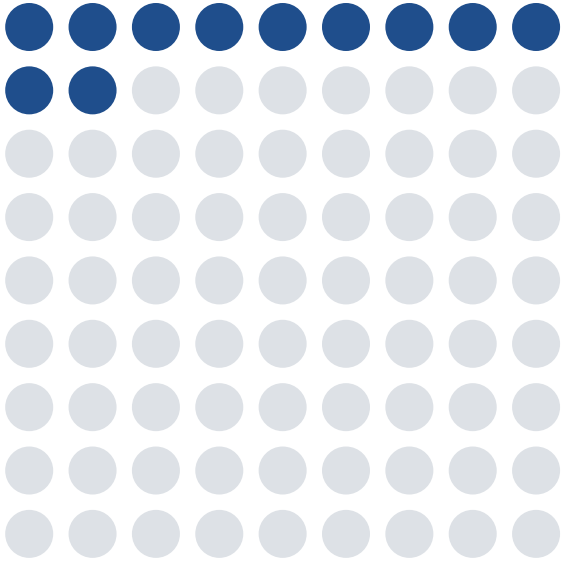
A lower aggregate perplexity does not establish better Afrikaans.

Does the decoder use the latent?



The decoder uses source-specific information in the latent.

Using the latent is not the same as keeping the content



11 / 81

passages reconstructed exactly



21 / 33

source numerals recovered

CONTENT CHANGED

Dienste en Armoedeverligting
→ **Ontwikkeling**
department renamed

20
→ **absent**
number lost

regulering
→ **betaling**
regulation → payment

Regeringsdepartement
→ **Regeringskommissie**
department → commission

Answers to the research questions

RQ1

Partly answered

Implementation control

English autoencoder learns to reconstruct

RQ2

Criterion not met

Representation fidelity

Uses the latent, but loses content

RQ3

Perplexity only

Comparative generation

Lower perplexity, unmatched baseline

Hypothesis not supported: reconstruction falls short, so a good perplexity score is not enough.

Contributions

1 **Larger denoiser, lower perplexity**

51.3 → 48.6 across five seeds; reconstruction unchanged.

2 **Uses the latent ≠ keeps the content**

chrF 68 vs 13 with the wrong latent, yet only 11 / 81 exact.

3 **Scores can hide lost meaning**

Good perplexity alongside lost names and numbers.

A better score is not the same as preserved meaning.